



MOSEK Licensing Guide
Release 10.0.44

MOSEK ApS

09 May 2023

Contents

1	Introduction	1
2	Contact Information	2
3	License Agreement	3
3.1	MOSEK end-user license agreement	3
3.2	Third party licenses	3
4	Quickstart	9
4.1	I don't have a license file yet	9
4.2	I have a license file	9
4.3	Local	9
4.4	Floating	10
5	License system basics	11
5.1	License Types	11
5.2	License File	12
5.3	Versions	14
5.4	Editing	14
6	Hostname and Hostid	15
6.1	Using MOSEK	15
6.2	Using FLEXlm	16
6.3	Using the OS tools	18
7	Floating license setup	19
7.1	Windows: Token server setup	20
7.2	Linux, macOS: Token server setup	27
7.3	Changing default ports and firewall and antivirus issues	30
7.4	Administration of the token server	31
8	Client setup	33
8.1	With a local license file	33
8.2	Without a local license file	34
8.3	Testing and debugging	34
9	Additional features	35
9.1	Options file	35
9.2	Borrowing a license	36
10	License in a Cloud Computing Environment	37
10.1	Example: Token server in Amazon EC2	37
11	Licensing FAQ	38
11.1	General questions	38
11.2	Floating license issues	40
11.3	Local file issues	42
11.4	Need more help	42

Chapter 1

Introduction

The **MOSEK** Optimization Suite is a commercial product that requires a valid license. This guide explains how the licensing system works and how to install a license.

Are you only interested in quick, simplified instructions? Jump directly to [Sec. 4](#).

Chapter 2

Contact Information

Phone	+45 7174 9373	
Website	mosek.com	
Email		
	sales@mosek.com	Sales, pricing, and licensing
	support@mosek.com	Technical support, questions and bug reports
	info@mosek.com	Everything else.
Mailing Address		
	MOSEK ApS	
	Fruebjergvej 3	
	Symbion Science Park, Box 16	
	2100 Copenhagen O	
	Denmark	

You can get in touch with **MOSEK** using popular social media as well:

Blogger	https://blog.mosek.com/
Google Group	https://groups.google.com/forum/#!forum/mosek
Twitter	https://twitter.com/mosektw
Linkedin	https://www.linkedin.com/company/mosek-aps
Youtube	https://www.youtube.com/channel/UCvIyectEVLp31NXeD5mIbEw

In particular **Twitter** is used for news, updates and release announcements.

Chapter 3

License Agreement

3.1 MOSEK end-user license agreement

Before using the **MOSEK** software, please read the license agreement available in the distribution at <MSKHOME>/mosek/10.0/mosek-eula.pdf or on the **MOSEK** website <https://mosek.com/products/license-agreement>. By using **MOSEK** you agree to the terms of that license agreement.

3.2 Third party licenses

MOSEK uses some third-party open-source libraries. Their license details follow.

zlib

MOSEK uses the *zlib* library obtained from the [zlib website](#). The license agreement for *zlib* is shown in [Listing 3.1](#).

Listing 3.1: *zlib* license.

```
zlib.h -- interface of the 'zlib' general purpose compression library
version 1.2.7, May 2nd, 2012

Copyright (C) 1995-2012 Jean-loup Gailly and Mark Adler

This software is provided 'as-is', without any express or implied
warranty. In no event will the authors be held liable for any damages
arising from the use of this software.

Permission is granted to anyone to use this software for any purpose,
including commercial applications, and to alter it and redistribute it
freely, subject to the following restrictions:

1. The origin of this software must not be misrepresented; you must not
   claim that you wrote the original software. If you use this software
   in a product, an acknowledgment in the product documentation would be
   appreciated but is not required.
2. Altered source versions must be plainly marked as such, and must not be
   misrepresented as being the original software.
3. This notice may not be removed or altered from any source distribution.

Jean-loup Gailly          Mark Adler
jloup@gzip.org            madler@alumni.caltech.edu
```

fplib

MOSEK uses the floating point formatting library developed by David M. Gay obtained from the [netlib website](#). The license agreement for *fplib* is shown in [Listing 3.2](#).

Listing 3.2: *fplib* license.

```
/*
 *
 * The author of this software is David M. Gay.
 *
 * Copyright (c) 1991, 2000, 2001 by Lucent Technologies.
 *
 * Permission to use, copy, modify, and distribute this software for any
 * purpose without fee is hereby granted, provided that this entire notice
 * is included in all copies of any software which is or includes a copy
 * or modification of this software and in all copies of the supporting
 * documentation for such software.
 *
 * THIS SOFTWARE IS BEING PROVIDED "AS IS", WITHOUT ANY EXPRESS OR IMPLIED
 * WARRANTY. IN PARTICULAR, NEITHER THE AUTHOR NOR LUCENT MAKES ANY
 * REPRESENTATION OR WARRANTY OF ANY KIND CONCERNING THE MERCHANTABILITY
 * OF THIS SOFTWARE OR ITS FITNESS FOR ANY PARTICULAR PURPOSE.
 *
 *****/
```

{fmt}

MOSEK uses the formatting library *{fmt}* developed by Victor Zverovich obtained from [github/fmt](#) and distributed under the MIT license. The license agreement for *{fmt}* is shown in [Listing 3.3](#).

Listing 3.3: *{fmt}* license.

```
Copyright (c) 2012 - present, Victor Zverovich

Permission is hereby granted, free of charge, to any person obtaining
a copy of this software and associated documentation files (the "Software"),
to deal in the Software without restriction, including without limitation
the rights to use, copy, modify, merge, publish, distribute, sublicense,
and/or sell copies of the Software, and to permit persons to whom the Software
is furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included
in all copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED,
INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR
A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE AUTHORS OR
COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY, WHETHER
IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM, OUT OF OR IN
CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE SOFTWARE.
```

Zstandard

MOSEK uses the *Zstandard* library developed by Facebook obtained from [github/zstd](https://github.com/facebook/zstd). The license agreement for *Zstandard* is shown in [Listing 3.4](#).

Listing 3.4: *Zstandard* license.

```
BSD License

For Zstandard software

Copyright (c) 2016-present, Facebook, Inc. All rights reserved.

Redistribution and use in source and binary forms, with or without modification,
are permitted provided that the following conditions are met:

* Redistributions of source code must retain the above copyright notice, this
  list of conditions and the following disclaimer.

* Redistributions in binary form must reproduce the above copyright notice,
  this list of conditions and the following disclaimer in the documentation
  and/or other materials provided with the distribution.

* Neither the name Facebook nor the names of its contributors may be used to
  endorse or promote products derived from this software without specific
  prior written permission.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND
ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED
WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE
DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT HOLDER OR CONTRIBUTORS BE LIABLE FOR
ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES
(INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES;
LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON
ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT
(INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS
SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.
```

OpenSSL

MOSEK uses the [LibReSSL](#) library, which is build on *OpenSSL*. *OpenSSL* is included under the *OpenSSL* license, [Listing 3.5](#), and the *LibReSSL* additions are licensed under the *ISC* license, [Listing 3.6](#).

Listing 3.5: *OpenSSL* license

```
=====
Copyright (c) 1998-2011 The OpenSSL Project. All rights reserved.

Redistribution and use in source and binary forms, with or without
modification, are permitted provided that the following conditions
are met:

1. Redistributions of source code must retain the above copyright
   notice, this list of conditions and the following disclaimer.

2. Redistributions in binary form must reproduce the above copyright
   notice, this list of conditions and the following disclaimer in
```

(continues on next page)

the documentation and/or other materials provided with the distribution.

3. All advertising materials mentioning features or use of this software must display the following acknowledgment:
"This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit. (<http://www.openssl.org/>)"
4. The names "OpenSSL Toolkit" and "OpenSSL Project" must not be used to endorse or promote products derived from this software without prior written permission. For written permission, please contact openssl-core@openssl.org.
5. Products derived from this software may not be called "OpenSSL" nor may "OpenSSL" appear in their names without prior written permission of the OpenSSL Project.
6. Redistributions of any form whatsoever must retain the following acknowledgment:
"This product includes software developed by the OpenSSL Project for use in the OpenSSL Toolkit (<http://www.openssl.org/>)"

THIS SOFTWARE IS PROVIDED BY THE OpenSSL PROJECT ``AS IS'' AND ANY EXPRESSED OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE OpenSSL PROJECT OR ITS CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

=====

This product includes cryptographic software written by Eric Young (ey@cryptsoft.com). This product includes software written by Tim Hudson (tjh@cryptsoft.com).

Listing 3.6: ISC license

Copyright (C) 1994-2017 Free Software Foundation, Inc.
Copyright (c) 2014 Jeremie Courreges-Anglas <jca@openbsd.org>
Copyright (c) 2014-2015 Joel Sing <jsing@openbsd.org>
Copyright (c) 2014 Ted Unangst <tedu@openbsd.org>
Copyright (c) 2015-2016 Bob Beck <beck@openbsd.org>
Copyright (c) 2015 Marko Kreen <markokr@gmail.com>
Copyright (c) 2015 Reyk Floeter <reyk@openbsd.org>
Copyright (c) 2016 Tobias Pape <tobias@netshed.de>

Permission to use, copy, modify, and/or distribute this software for any purpose with or without fee is hereby granted, provided that the above copyright notice and this permission notice appear in all copies.

(continued from previous page)

```
THE SOFTWARE IS PROVIDED "AS IS" AND THE AUTHOR DISCLAIMS ALL
WARRANTIES WITH REGARD TO THIS SOFTWARE INCLUDING ALL IMPLIED
WARRANTIES OF MERCHANTABILITY AND FITNESS. IN NO EVENT SHALL THE
AUTHOR BE LIABLE FOR ANY SPECIAL, DIRECT, INDIRECT, OR CONSEQUENTIAL
DAMAGES OR ANY DAMAGES WHATSOEVER RESULTING FROM LOSS OF USE, DATA OR
PROFITS, WHETHER IN AN ACTION OF CONTRACT, NEGLIGENCE OR OTHER
TORTIOUS ACTION, ARISING OUT OF OR IN CONNECTION WITH THE USE OR
PERFORMANCE OF THIS SOFTWARE.
```

mimalloc

MOSEK uses the *mimalloc* memory allocator library from [github/mimalloc](https://github.com/mimalloc). The license agreement for *mimalloc* is shown in [Listing 3.7](#).

Listing 3.7: *mimalloc* license.

```
MIT License

Copyright (c) 2019 Microsoft Corporation, Daan Leijen

Permission is hereby granted, free of charge, to any person obtaining a copy
of this software and associated documentation files (the "Software"), to deal
in the Software without restriction, including without limitation the rights
to use, copy, modify, merge, publish, distribute, sublicense, and/or sell
copies of the Software, and to permit persons to whom the Software is
furnished to do so, subject to the following conditions:

The above copyright notice and this permission notice shall be included in all
copies or substantial portions of the Software.

THE SOFTWARE IS PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR
IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY,
FITNESS FOR A PARTICULAR PURPOSE AND NONINFRINGEMENT. IN NO EVENT SHALL THE
AUTHORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER
LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT OR OTHERWISE, ARISING FROM,
OUT OF OR IN CONNECTION WITH THE SOFTWARE OR THE USE OR OTHER DEALINGS IN THE
SOFTWARE.
```

BLASFEO

MOSEK uses the *BLASFEO* linear algebra library developed by Gianluca Frison, obtained from [github/blasfeo](https://github.com/blasfeo). The license agreement for *BLASFEO* is shown in [Listing 3.8](#).

Listing 3.8: *blasfeo* license.

```
BLASFEO -- BLAS For Embedded Optimization.
Copyright (C) 2019 by Gianluca Frison.
Developed at IMTEK (University of Freiburg) under the supervision of Moritz Diehl.
All rights reserved.

The 2-Clause BSD License

Redistribution and use in source and binary forms, with or without
modification, are permitted provided that the following conditions are met:

1. Redistributions of source code must retain the above copyright notice, this
```

(continues on next page)

list of conditions and the following disclaimer.

2. Redistributions in binary form must reproduce the above copyright notice, this list of conditions and the following disclaimer in the documentation and/or other materials provided with the distribution.

THIS SOFTWARE IS PROVIDED BY THE COPYRIGHT HOLDERS AND CONTRIBUTORS "AS IS" AND ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE ARE DISCLAIMED. IN NO EVENT SHALL THE COPYRIGHT OWNER OR CONTRIBUTORS BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

oneTBB

MOSEK uses the *oneTBB* parallelization library which is part of *oneAPI* developed by Intel, obtained from [github/oneTBB](https://github.com/oneTBB), licensed under the Apache License 2.0. The license agreement for *oneTBB* can be found in <https://github.com/oneapi-src/oneTBB/blob/master/LICENSE.txt> .

Chapter 4

Quickstart

These are quickstart instructions for users who just want the most basic, working, easy and quick license setup and do not care about the more advanced configuration features.

4.1 I don't have a license file yet

Free licenses

- To obtain a **trial license** go to <https://www.mosek.com/products/trial/>
- To obtain a **personal academic license** go to <https://www.mosek.com/products/academic-licenses/>
- To obtain an **institutional academic license** go to <https://www.mosek.com/products/academic-licenses/>
- If you have a **custom license** go to <https://www.mosek.com/license/request/custom/> and enter the code you received.

Commercial licenses

Assuming you purchased a product (<https://www.mosek.com/sales/order/>) you will obtain a license file from us. For a **floating license** or **server (node-locked) license** you will first have to follow [Sec. 6](#) to determine the **hostname** and **hostid** of the designated machine. Contact us at license@mosek.com.

4.2 I have a license file

- Do you have a **trial license** or **personal academic license** or **server (node-locked) license** or **group license** or **custom license**? Go to [Sec. 4.3](#).
- Do you have a **floating license** or **institutional academic license**? Go to [Sec. 4.4](#).

4.3 Local

Put your license file in:

<code>\$HOME/mosek/mosek.lic</code>	(Linux/OSX)
<code>%USERPROFILE%\mosek\mosek.lic</code>	(Windows)

In most cases that is:

<code>/home/myusername/mosek/mosek.lic</code>	(Linux)
<code>/Users/myusername/mosek/mosek.lic</code>	(OSX)
<code>C:\Users\myusername\mosek\mosek.lic</code>	(Windows)

If that folder does not exist (which it most likely does not if you are using **MOSEK** for the first time), then create it.

Restart any software using **MOSEK** (for example MATLAB, R, etc.) if you updated an existing license.

Need more options or are there issues? See [Sec. 8](#).

4.4 Floating

- Start (or restart) the token server using the instructions in [Sec. 7](#).
- On each user machine where you will run **MOSEK** do exactly the same as in [Sec. 4.3](#).

Are there network issues? See [Sec. 7.3](#).

Chapter 5

License system basics

The **MOSEK** Optimization Suite is licensed software which means a valid license is required. A license is provided by a license file that specifies:

- which features in **MOSEK** have been licensed (an example of a feature is the nonlinear extension PTON),
- how many copies of a feature can be used simultaneously,
- an expiration date of each feature,
- for floating licenses, the identifier of the server the license is tied to.

5.1 License Types

The license is managed by the FLEXlm (<http://www.flexerasoftware.com/>) license manager included in **MOSEK**. FLEXlm has two types of licenses:

- **floating**: license tied to a particular computer that acts as a *token server*. **MOSEK** can be used on any computer connected to the token server through the local area network (LAN). In particular **MOSEK** can be used on the token server itself. Setting up a floating license is described in [Sec. 7](#).
- **server** (also known as **node-locked**): license tied to a particular computer that allows unlimited use of the licensed features on that particular machine. Setting up a server license is described in [Sec. 8](#).

Moreover, note that:

- **institutional academic** licenses are floating licenses.
- **trial**, **group** and **personal academic** licenses behave as server licenses, except that they are not tied to a specific computer but can be used on any machine where the license file is present.

For floating, server and institutional academic licenses some computer-dependent information must be provided:

- **hostname**: the name that identifies the computer in the network,
- **hostid**: a unique computer identifier (typically its MAC address).

Instructions for obtaining hostname and hostid can be found in [Sec. 6](#).

5.2 License File

A license file is a plain text file that can be opened for inspection using any plain text editor (such as vim or emacs on Linux, or notepad on Windows). It is sometimes useful to inspect the file to check the expiration date, the activated features and computer information.

5.2.1 Floating license file

Listing 5.1: An example of license file for a floating license.

```
#
# MOSEK License File (LIF-1234)
# https://www.mosek.com
#

SERVER hulk f4ed3061a731
USE_SERVER
VENDOR MOSEKLM
FEATURE PTS MOSEKLM 10 permanent 5 VENDOR_STRING=-1*-1*-1*0 \
    BORROW=744 SN=LIF-1234 SIGN="0123 4567 89AB CDEF 0123 4567 89AB \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB" \
    SIGN2="0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF"
FEATURE PTON MOSEKLM 10 permanent 5 VENDOR_STRING=-1*-1*-1*0 \
    BORROW=744 SN=LIF-1234 SIGN="0123 4567 89AB CDEF 0123 4567 89AB \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB" \
    SIGN2="0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123 4567 89AB"
```

Listing 5.1 shows an extract of a license file for a floating license. Lines starting from # are comments. The syntax is (obligatory fields in <> and optional fields in []):

Listing 5.2: Syntax of a floating license file.

```
SERVER <hostname> <hostid> [lmgrd_port]
USE_SERVER
VENDOR MOSEKLM [vendor_options] [port=moseklm_port]
FEATURE <feature_name> MOSEKLM <mosek_version> <expiry_date> <number_of_tokens>␣
↪VENDOR_STRING=-1*-1*-1*0 \
    ... private data ...
FEATURE <feature_name> MOSEKLM <mosek_version> <expiry_date> <number_of_tokens>␣
↪VENDOR_STRING=-1*-1*-1*0 \
    ... private data ...
```

There is one feature entry per each feature the license file supports (PTS, PTON or others).

5.2.2 Server (node-locked) or unrestricted license file

Listing 5.3: An example of license file for a server (node-locked) license.

```
#
# MOSEK License File (LIF-1234)
# https://www.mosek.com
#

VENDOR MOSEKLM
FEATURE PTS MOSEKLM 10 permanent uncounted VENDOR_STRING=-1*-1*-1*0 \
    HOSTID=f4ed3061a731 BORROW=744 SN=LIF-1234 TS_OK SIGN="0123 \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123" SIGN2="0123 4567 89AB CDEF 0123 4567\
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF"
FEATURE PTON MOSEKLM 10 permanent uncounted VENDOR_STRING=-1*-1*-1*0 \
    HOSTID=f4ed3061a731 BORROW=744 SN=LIF-1234 TS_OK SIGN="0123 \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123" SIGN2="0123 4567 89AB CDEF 0123 4567\
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF \
    0123 4567 89AB CDEF 0123 4567 89AB CDEF 0123 4567 89AB CDEF"
```

Listing 5.3 shows an extract of a license file for a floating license. In case of unrestricted (trial, personal academic) licenses the <hostid> will be ANY. Lines starting from # are comments. The syntax is (obligatory fields in <> and optional fields in []):

Listing 5.4: Syntax of a node-locked license file.

```
VENDOR MOSEKLM
FEATURE <feature_name> MOSEKLM <mosek_version> <expiry_date> uncounted VENDOR_STRING==
↪1*-1*-1*0 \
    HOSTID=<hostid>    ... private data ...
FEATURE <feature_name> MOSEKLM <mosek_version> <expiry_date> uncounted VENDOR_STRING==
↪1*-1*-1*0 \
    HOSTID=<hostid>    ... private data ...
```

There is one feature entry per each feature the license file supports (PTS, PTON or others).

5.3 Versions

- **MOSEK** version 10 requires a license file with version at least 10.
- The license file for **MOSEK** version 10 will support all **MOSEK** clients with versions less than or equal to 10.
- Only the major version matters.
- For floating licenses, use the token server from the same, or newer **MOSEK** distribution as the one you are using in the client. Token server binaries should be as new as the newest client contacting the token server. When the FLEXlm version used by the token server falls too far behind the version used in the **MOSEK** client, they will no longer be able to communicate.
- If you are using your own `lmgrd` for floating licenses then its version must be at least as new as that used by the newest **MOSEK** client contacting the server.
- The FLEXlm version used by **MOSEK** can be found by running `mosek -f` in the command-line tool. The history of recent FLEXlm upgrades in **MOSEK** is as follows:

Mosek version	FlexLM version
from 10.0.0	11.18.3
from 9.3.0	11.18.1
from 9.1.5	11.16.4
from 8.0	11.14.1

5.4 Editing

The user may edit the sections `lmgrd_port`, `port=moseklm_port` and `vendor_options` in the license files. Other edits may invalidate the license file. Good reasons for such edits exist only for floating license files and include:

- changing/fixing the port numbers used for floating license checkouts. See [Sec. 7.3](#).
- attaching an options file, for instance to reserve a number of floating licenses for specific users or groups. See [Sec. 9.1](#)

Otherwise it should not be necessary to edit the file.

Chapter 6

Hostname and Hostid

The **hostname** and **hostid** are the two basic computer identifiers used in some of **MOSEK** license files.

- **hostname** is just the standard host name of the machine.
- **hostid** is, in most cases, the MAC address of a network card, or of one of the network cards. It is a sequence of 12 hexadecimal digits. There is no need for the card to have an active connection.

Both identifiers can be obtained in various ways, either with **MOSEK** or standard tools available in the operating system. We outline these below.

Important: A modern machine will often have many network interfaces, some physical (hardware network devices) and possibly also some virtual. The licensing software will detect all of them and present them as a list in no particular order, but we only need ONE of these MAC addresses as a **hostid** the license will be bound to.

In such cases please indicate which of the list of MAC addresses is the preferred choice for **hostid**. It is recommended to choose one that corresponds to one of permanent physical network cards. That assures the interface will not easily disappear, which would render the license file invalid. You can explore the network devices in your system settings or use the tools described in [Sec. 6.3](#) to match MACs to actual interfaces.

6.1 Using MOSEK

6.1.1 Command line

In the directory with **MOSEK** binaries (<MSKHOME>/mosek/10.0/tools/platform/<PLATFORM>/bin/) run the command

```
mosek -f
```

It will produce output similar to

```
MOSEK Version 10.0.34 (Build date: 2023-1-6 13:40:00)
Copyright (c) MOSEK ApS, Denmark WWW: mosek.com
Platform: Linux/64-X86

FlexLM
Version           : 11.18
Hostname          : hulk
Host ID           : "f4ed3061a731 00aa720b7fa"
License path      : /home/username/mosek/mosek.lic

Operating system variables
LD_LIBRARY_PATH   :
```

(continues on next page)

(continued from previous page)

```
*** No input file specified. No optimization is performed.
```

```
Return code - 0 [MSK_RES_OK]
```

Both `hostname` and a list of possible `hostid` are present in the output.

6.1.2 From an API

If you only installed **MOSEK** as a client package (for example via `pip`, `Anaconda` or `NuGet`) and do not have the full distribution then you can get the same output by calling the `echointro` method of the Optimizer API environment object `Env`. Note, however, that for floating licenses you will still have to download the full **MOSEK** distribution package to obtain the license server binaries.

Here is an example of this in Python:

```
import mosek, sys

env = mosek.Env()
env.set_Stream(mosek.streamtype.log, sys.stdout.write)
env.echointro(1)
```

6.2 Using FLEXlm

6.2.1 FLEXlm command line

In the directory with **MOSEK** binaries (`<MSKHOME>/mosek/10.0/tools/platform/<PLATFORM>/bin/`) run the command

```
./lmutil lmhostid          (Linux/OSX)
lmutil lmhostid            (Windows)
```

This will give you the list of possible `hostid` similar to:

```
lmutil - Copyright (c) 1989-2021 Flexera. All Rights Reserved.
The FlexNet host ID of this machine is "f4ed3061a731 00aa720b7fa"
Only use ONE from the list of hostids.
```

The `hostname` is obtained simply by running

```
hostname
```

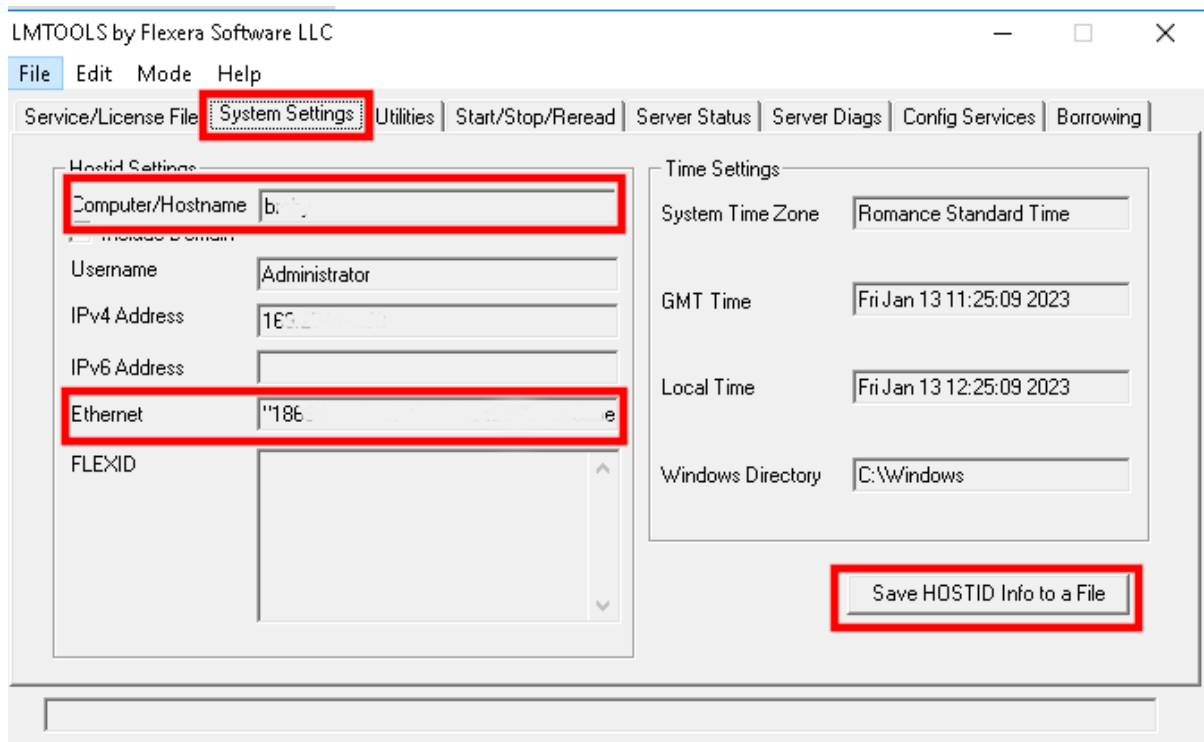
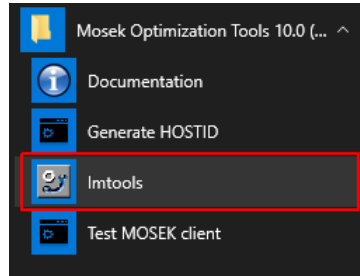
Linux: To use the floating license server or run `lmutil` the *Linux standard base 3.0* must be installed. This package is called `lsb-base` or `lsb` in most Linux distributions. Without it you can expect an error similar to:

```
./lmutil: No such file or directory
```

6.2.2 LMTOOLS graphical utility (Windows)

Start the **LMTOOLS** graphical utility, either from the **MOSEK** Start menu or from `<MSKHOME>/mosek/10.0/tools/platform/<PLATFORM>/bin/`.

Navigate to the tab **System settings**. The `hostname` and list of `hostid` are shown in the **Hostid Settings** panel under, respectively, *Computer/Hostname* and *Ethernet*. You can also choose to save all information to a file with the button in the bottom-right corner.



6.3 Using the OS tools

6.3.1 Command line

Start the terminal or command prompt. On all platforms the `hostname` can be obtained by running the command

```
hostname
```

To obtain the `hostid` generate a list of network interfaces by running one of the following commands. Some administrator permissions may be required.

<code>ifconfig</code>	(Linux/OSX)
<code>ip addr</code>	(Linux)
<code>netstat -i</code>	(Linux/OSX)
<code>wmic nic get AdapterType, Name, Installed, MACAddress</code>	(Windows)
<code>ipconfig /all</code>	(Windows)
<code>getmac /v</code>	(Windows)

In each case you will obtain a list of network interfaces with more or less detailed descriptions. Identify one corresponding to the device you want to bind to and use its MAC address as `hostid` (it will usually be displayed in the format `01:23:45:67:90:ab` or similar and referred to as `MAC`, `ether`, `Ethernet` or similar). If in doubt save the output of the command to a file and send us the file.

Important: Linux: DO NOT use the command line tool `hostid`. It generates a different identifier with 8 hexadecimal digits which we cannot use.

6.3.2 System settings

In the system settings find the section with network devices, find a network card and browse through its details to find the MAC address. We omit the details as they vary between operating systems.

Chapter 7

Floating license setup

A **floating license** is tied to a particular computer acting as a *token server*. A token server is a *service* on Windows and a *daemon* on UNIX that serves license tokens to **MOSEK** client programs over the LAN.

You may think of the token server as a computer with a bag of license tokens. Whenever a client computer starts using **MOSEK**, a license token is requested from the token server, and when **MOSEK** completes it sends back the license token to the token server. The following diagram Fig. 7.1 conveys the overall idea.

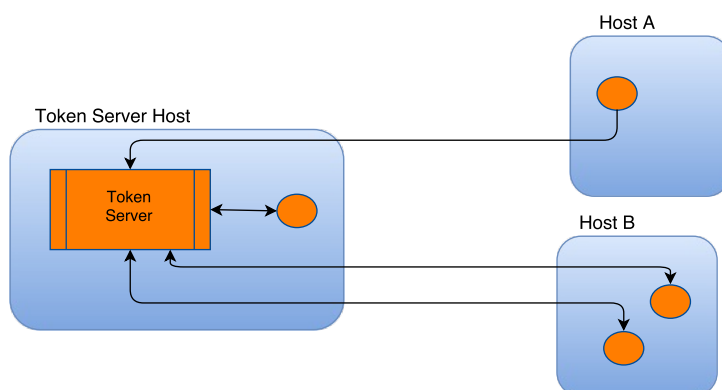


Fig. 7.1: General floating license scheme: any **MOSEK** instances that can connect to the token server can get a valid license.

This implies that you cannot use more license tokens than are available at any given point in time. Moreover, **MOSEK** can be used on any computer connected to the token server through the local area network. In particular **MOSEK** can also be used on the token server itself.

Note that

- a floating license file starts with the word **SERVER**. The format is described in Sec. 5.2.1. Installing a license file without a **SERVER** line with a token server is *NOT* needed and is *NOT* possible.
- at most one token server can be running on any given machine.
- The token server consists of two daemons: **lmgrd** - the token server daemon running as a service, and **MOSEKLM** - a vendor daemon started by **lmgrd** specifically for handling **MOSEK** licenses.

The following subsections guide through the setup of a token server on Windows, Linux and macOS, and later discuss restarting and upgrading the server.

7.1 Windows: Token server setup

7.1.1 Prerequisites and locating the files

- Make sure you have **administrative privileges**.
- Locate the **bin** folder of your **MOSEK** installation, that is

<MSKHOME>\mosek\10.0\tools\platform\<PLATFORM>\bin\

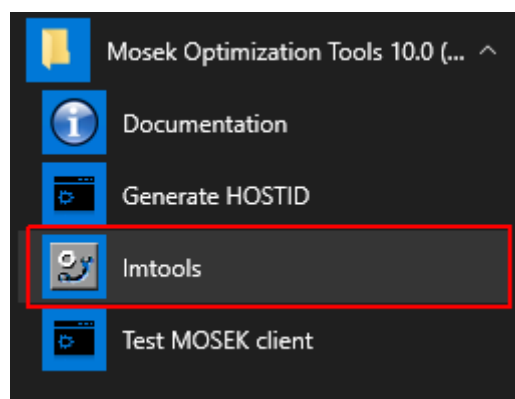
where <PLATFORM> is the platform you are working on. That folder contains the files **lmgrd.exe** and **MOSEKLM.exe** required by the license system.

This PC > Local Disk (C:) > Program Files > Mosek > 10.0 > tools > platform > win64x86 > bin				
Name	Date modified	Type	Size	
fusion64_10_0.lib	2/25/2022 12:54 PM	LIB File	23,940 KB	
installs	2/25/2022 12:54 PM	Application	412 KB	
libmosek64_10_0.a	2/25/2022 12:54 PM	A File	387 KB	
lmgrd	2/25/2022 12:54 PM	Application	1,185 KB	
lmtools	2/25/2022 12:54 PM	Application	1,411 KB	
lmutil	2/25/2022 12:54 PM	Application	1,244 KB	
mosek	2/25/2022 12:57 PM	Application	306 KB	
mosek.jar	2/25/2022 12:57 PM	JAR File	421 KB	
mosek64_10_0.dll	2/25/2022 12:57 PM	Application extens...	21,972 KB	
mosek64_10_0.lib	2/25/2022 12:57 PM	LIB File	107 KB	
mosekcli	2/25/2022 12:57 PM	Application	72 KB	
mosekconsole.py	2/25/2022 12:57 PM	PY File	49 KB	
mosekdotnet.dll	2/25/2022 12:58 PM	Application extens...	719 KB	
mosekdotnet10_0.dll	2/25/2022 12:58 PM	Application extens...	719 KB	
mosekjava10_0.dll	2/25/2022 12:58 PM	Application extens...	273 KB	
MOSEKLM	2/25/2022 12:58 PM	Application	2,494 KB	

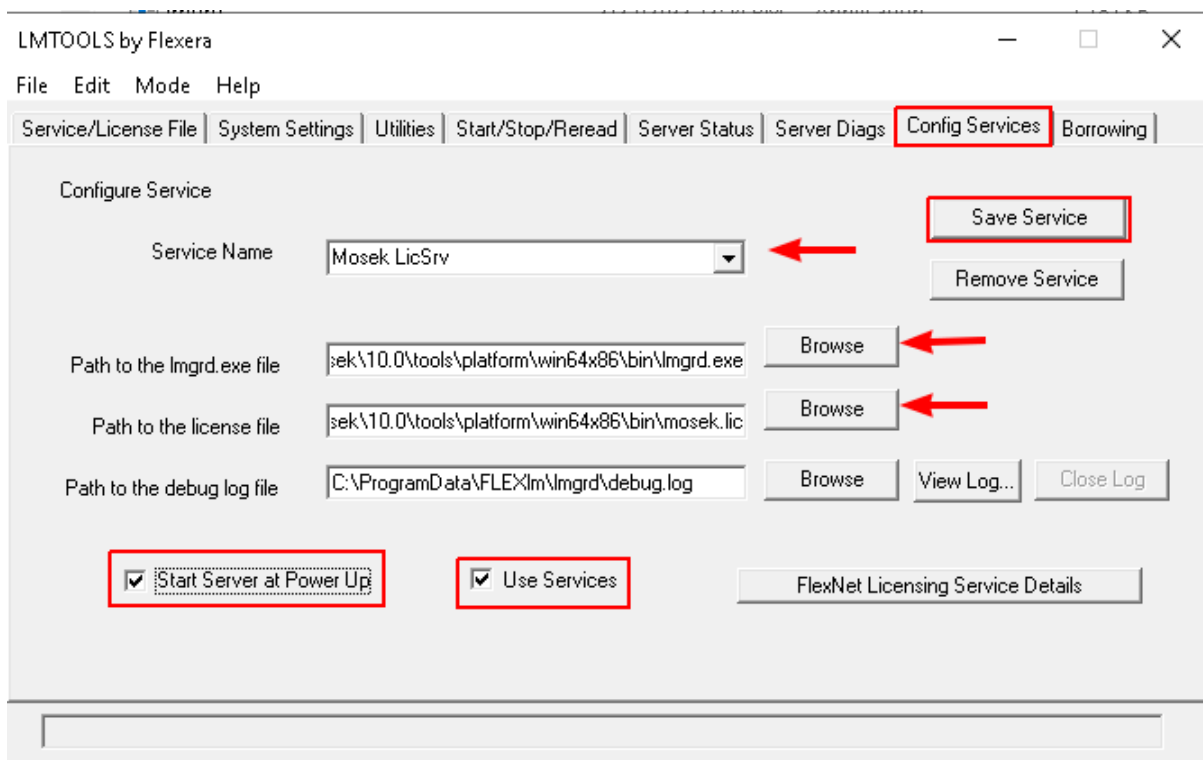
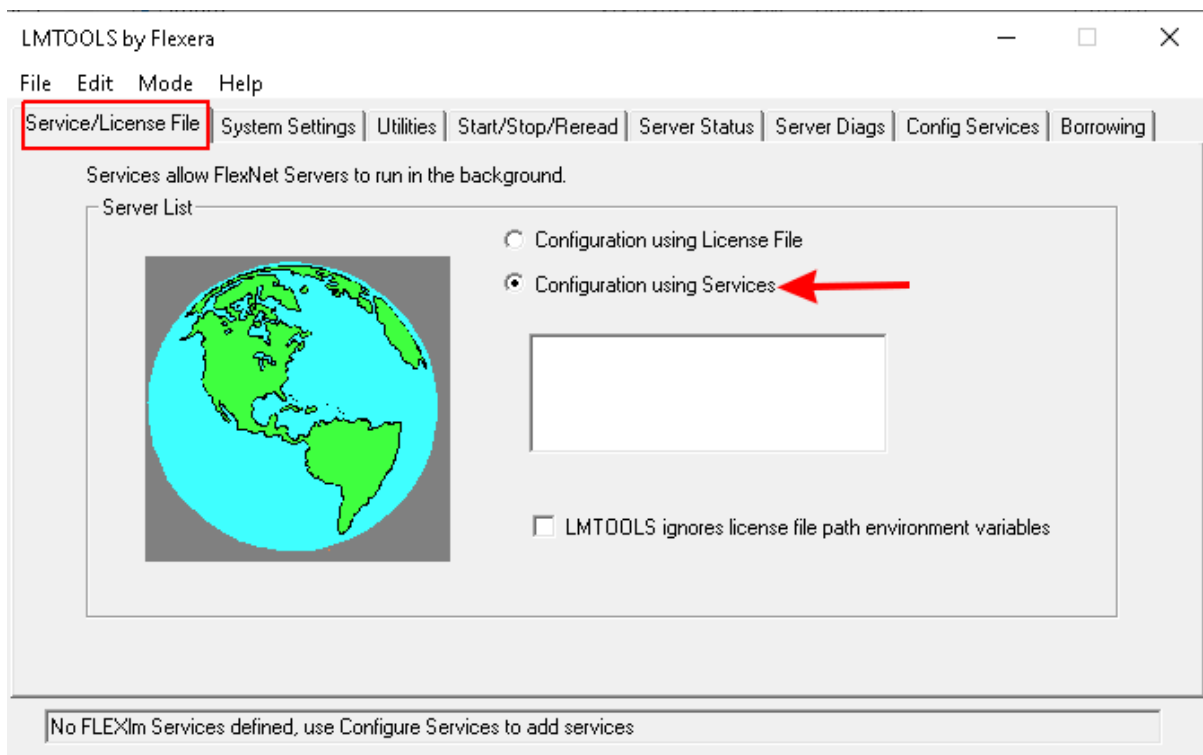
- Download the license file and store it on the local drive.

7.1.2 Installation of the service

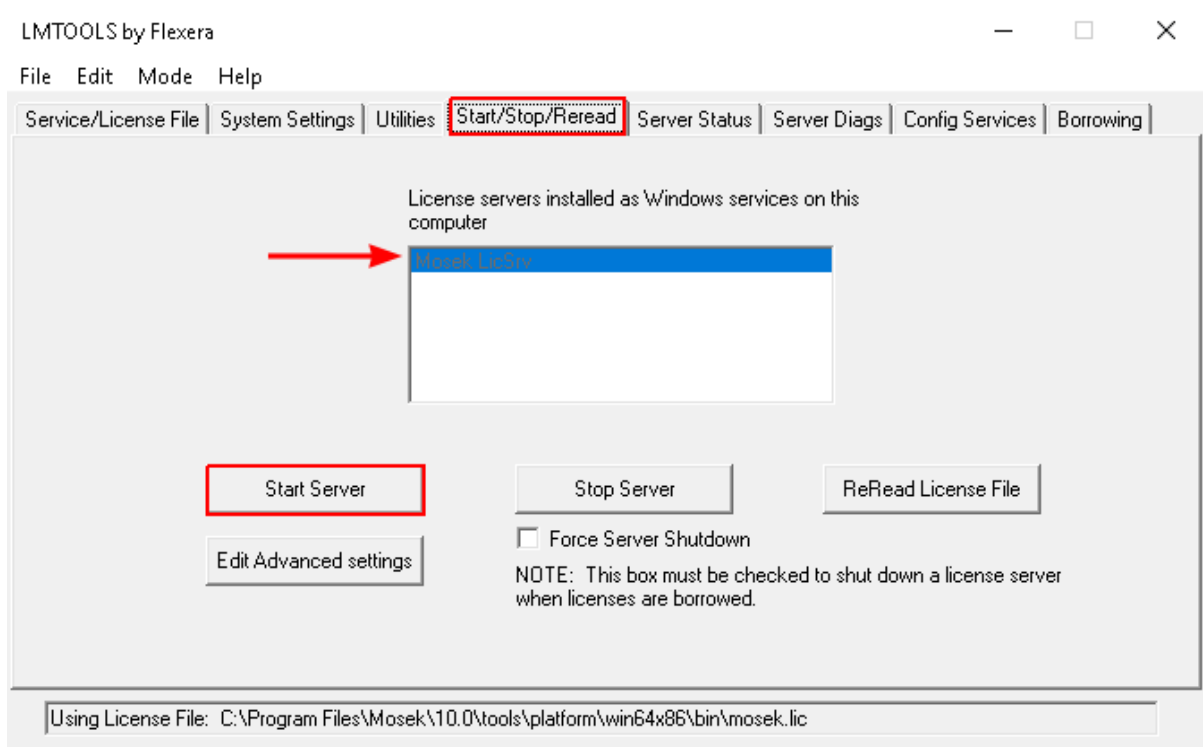
- Locate and start the program **lmtools**. You can find it in the same **bin** folder as above and also in the **MOSEK** Start Menu if **MOSEK** was installed with the MSI installer.



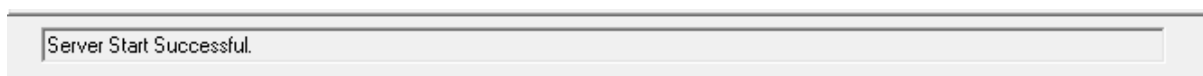
- In the **Service/License** file TAB select **Configuration using Services**.



- Go to the **Config Services** TAB.
 - In *Service name* input your chosen name for the service.
 - In *Path to the lmgrd.exe file* navigate to the previously located **lmgrd.exe** file in the **bin** folder of the **MOSEK** installation (or another location if you moved it).
 - In *Path to the license file* navigate to the location of your **MOSEK** license file.
 - Tick the box *Use Services* and then tick the box *Start server at Power Up*.
 - Click *Save Service* and confirm that you want to save the service.
- Go to the **Start/Stop/Reread** TAB.



- Choose your service from the list and click *Start Server*.
- You should see the message *Server start successful* in the status line at the bottom.



7.1.3 Checking that startup succeeded

To get a quick overview of the server start go to the **Config Services** TAB and click *View log*. A log file from a successful start should look similarly to:

```
9:49:15 (lmgrd) -----
9:49:15 (lmgrd)   Please Note:
9:49:15 (lmgrd)
9:49:15 (lmgrd)   This log is intended for debug purposes only.

.....
```

(continues on next page)

(continued from previous page)

```
9:49:15 (lmgrd)
9:49:15 (lmgrd) -----
9:49:15 (lmgrd)
9:49:15 (lmgrd)
9:49:15 (lmgrd) Server's System Date and Time: Mon Feb 28 2022 09:49:15 CET
9:49:15 (lmgrd) SLOG: Summary LOG statistics is enabled.
9:49:15 (lmgrd) FlexNet Licensing (v11.18.3.0 build 283040 x64_lsb) started on ↵
↵hostname (linux) (2/28/2022)
9:49:15 (lmgrd) Copyright (c) 1988-2021 Flexera. All Rights Reserved.
9:49:15 (lmgrd) World Wide Web: http://www.flexerasoftware.com
9:49:15 (lmgrd) License file(s): /home/username/mosek/mosek-floating.lic
9:49:15 (lmgrd) lmgrd tcp-port 27007
9:49:15 (lmgrd) (@lmgrd-SLOG@) =====
9:49:15 (lmgrd) (@lmgrd-SLOG@) === LMGRD ===
9:49:15 (lmgrd) (@lmgrd-SLOG@) Start-Date: Mon Feb 28 2022 09:49:15 CET
9:49:15 (lmgrd) (@lmgrd-SLOG@) PID: 2147425
9:49:15 (lmgrd) (@lmgrd-SLOG@) LMGRD Version: v11.18.3.0 build 283040 x64_lsb ( build ↵
↵283040 (ipv6))
9:49:15 (lmgrd) (@lmgrd-SLOG@)
9:49:15 (lmgrd) (@lmgrd-SLOG@) === Network Info ===
9:49:15 (lmgrd) (@lmgrd-SLOG@) Listening port: 27007
9:49:15 (lmgrd) (@lmgrd-SLOG@)
9:49:15 (lmgrd) (@lmgrd-SLOG@) === Startup Info ===
9:49:15 (lmgrd) (@lmgrd-SLOG@) Server Configuration: Single Server
9:49:15 (lmgrd) (@lmgrd-SLOG@) Command-line options used at LS startup: -c /home/
↵username/mosek/mosek-floating.lic
9:49:15 (lmgrd) (@lmgrd-SLOG@) License file(s) used: /home/username/mosek/mosek-
↵floating.lic
9:49:15 (lmgrd) (@lmgrd-SLOG@) =====
9:49:15 (lmgrd) Starting vendor daemons ...
9:49:15 (lmgrd) Started MOSEKLM (internet tcp_port 36867 pid 2147427)
9:49:15 (MOSEKLM) FlexNet Licensing version v11.18.3.0 build 283040 x64_lsb
9:49:15 (MOSEKLM) SLOG: Summary LOG statistics is enabled.
9:49:15 (MOSEKLM) SLOG: FNPLS-INTERNAL-CKPT1
9:49:15 (MOSEKLM) SLOG: VM Status: 0
9:49:15 (MOSEKLM) SLOG: FNPLS-INTERNAL-CKPT5
9:49:15 (MOSEKLM) SLOG: TPM Status: 0
9:49:15 (MOSEKLM) SLOG: FNPLS-INTERNAL-CKPT6
9:49:15 (MOSEKLM) Server started on hostname for: PTS
9:49:15 (MOSEKLM) PTON
9:49:15 (MOSEKLM) EXTERNAL FILTERS are OFF
9:49:15 (lmgrd) MOSEKLM using TCP-port 36867
9:49:15 (MOSEKLM) SLOG: Statistics Log Frequency is 240 minute(s).
9:49:15 (MOSEKLM) SLOG: TS update poll interval is 600 seconds.
9:49:15 (MOSEKLM) SLOG: Activation borrow reclaim percentage is 0.
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) =====
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) === Vendor Daemon ===
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Vendor daemon: MOSEKLM
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Start-Date: Mon Feb 28 2022 09:49:15 CET
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) PID: 2147427
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) VD Version: v11.18.3.0 build 283040 x64_lsb ( ↵
↵build 283040 (ipv6))
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@)
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) === Startup/Restart Info ===
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Options file used: None
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Is vendor daemon a CVD: No
```

(continues on next page)

(continued from previous page)

```
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Is FlexNet Licensing Service installed and
↪compatible: No
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) FlexNet Licensing Service Version: -NA-
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Is TS accessed: No
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) TS access time: -NA-
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Number of VD restarts since LS startup: 0
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@)
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) === Network Info ===
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Listening port: 36867
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Daemon select timeout (in seconds): 1
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@)
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) === Host Info ===
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Host used in license file: hostname
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) HostID node-locked in license file: b47acd81a33c
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) HostID of the License Server: "b47acd81a33c
↪b47acd81a33d"
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Running on Hypervisor: Not determined - treat as
↪Physical
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) =====
9:49:15 (MOSEKLM) TCP_NODELAY NOT enabled
9:49:15 (MOSEKLM) Listener Thread: running
9:49:15 (MOSEKLM) Starting diagnostics port listener thread (DPLT)
9:49:15 (MOSEKLM) Starting diagnostics output thread (DRQT)
```

In particular, the major information contained in the file is:

- the port `lmgrd` is running on (in this case 27007, the default port used by **MOSEK**),
- the port `MOSEKLM` is running on (in this case 36867, by default a value assigned dynamically at startup),
- the path to the license file that was used,
- the hostname and hostid of the machine,
- the licensed part IDs (in this case `PTS` and `PTON`)

If the log file indicates errors consult the next rubric and [Sec. 11.2](#).

7.1.4 Troubleshooting possible permission issues.

Here we mention one Windows-specific permission-related error which can prevent the startup. It manifests as follows: the startup takes quite long, ends in a message such as:



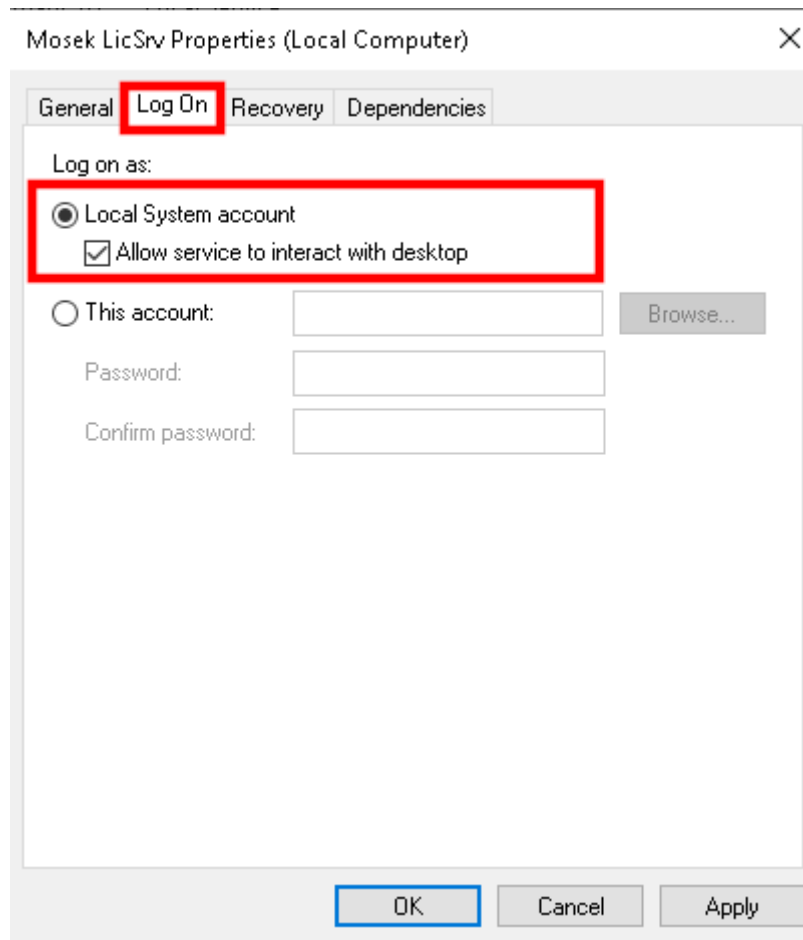
VD is starting, please check vendor daemon's status in debug log

and the log file looks similarly to:

```
6:02:10 (lmgrd) license manager: can't initialize:Cannot find license file.
6:02:10 (lmgrd) The license files (or license server system network addresses)
↪attempted are
6:02:10 (lmgrd) listed below. Use LM_LICENSE_FILE to use a different license file,
6:02:10 (lmgrd) or contact your software provider for a license file.
6:02:10 (lmgrd) Filename: "C:\flexlm\license.dat"
6:02:10 (lmgrd) License Path: "C:\flexlm\license.dat"
6:02:10 (lmgrd) FlexNet Licensing error:-1,359
6:02:10 (lmgrd) System Error:2 No such file or directory
6:02:10 (lmgrd) Using license file "C:\flexlm\license.dat"
```

Then you may be experiencing an issue with service permissions.

- **Recommended solution.** Open the *Windows Administrative Tools*, navigate to *Services (Local)* and find the service with the name you just created. Open the configuration for that service and in the **Log On** TAB choose *Local System account* and tick the checkbox for *Allow service to interact with desktop*:



Save the new configuration and start the service again in the **Start/Stop/Reread** TAB of *lmtools*.

- **Alternative solution.** Move the license file and the files *lmgrd.exe* and *MOSEKLM.exe* all three to the same new folder. Repeat the configuration, this time pointing *lmtools* to the paths in this common location.

7.1.5 Testing the Token Server

In order to verify that the token server is running, go to the **Server Status** TAB and click *Perform Status Enquiry*.

The result for a successfully running token server should be similar to the one below:

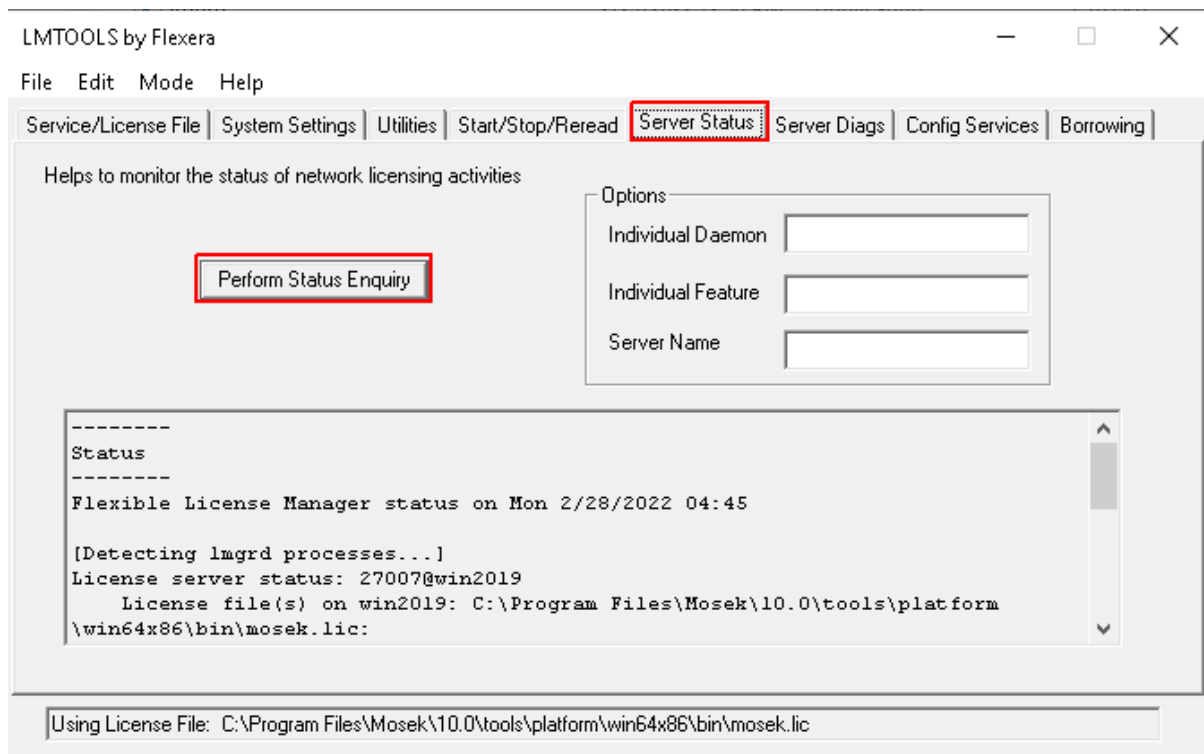
```
lmutil - Copyright (c) 1989-2021 Flexera. All Rights Reserved.
Flexible License Manager status on Mon 2/28/2022 10:09

License server status: 27007@hostname
  License file(s) on hostname: /home/username/mosek/mosek-floating.lic:

  hostname: license server UP (MASTER) v11.18.3

Vendor daemon status (on hostname):
```

(continues on next page)



(continued from previous page)

```
MOSEKLM: UP v11.18.3
Feature usage info:

Users of PTS:  (Total of 5 licenses issued;  Total of 0 licenses in use)

Users of PTON: (Total of 5 licenses issued;  Total of 0 licenses in use)
```

7.1.6 Troubleshooting

If the server won't start, or the client can't check out the license first check [Sec. 11.2](#). It covers most typical issues. Always consult the log file `lmgrd.log` of the server.

If that doesn't help please contact **MOSEK** support at support@mosek.com. Please include the client error messages and the server log file `lmgrd.log`. See [Sec. 11.4](#).

7.2 Linux, macOS: Token server setup

Below is a step-by-step guide to starting the token server on Linux and macOS. The same instructions can also be applied to start the token server on Windows from the command line, although for most Windows users the approach from [Sec. 7.1](#), which installs the token server as a service, is preferable.

7.2.1 Prerequisites and locating the files

On Linux *Linux Standard Base* (LSB) 3.0 or later must be installed in order for token server to work. The LSB package is called `lsb-base` or `lsb` in most Linux distributions.

The programs `lmgrd` and `MOSEKLM` required for installation can be found in the `bin` folder inside the **MOSEK** installation, that is:

```
<MSKHOME>/mosek/10.0/tools/platform/<PLATFORM>/bin/
```

where `<PLATFORM>` is the platform you are working on.

Starting the token server

To start the token server open the terminal, go to the `bin` folder of the **MOSEK** installation mentioned above and run:

```
./lmgrd -c PATH_TO_LICENSE -l lmgrd.log
```

where `PATH_TO_LICENSE` is the path to your license file. The token server will save a log file in the location given by the `-l` command line parameter.

7.2.2 Checking that startup succeeded

If the token server was started successfully the `lmgrd.log` file will look similar to this

```
9:49:15 (lmgrd) -----
9:49:15 (lmgrd)   Please Note:
9:49:15 (lmgrd)
9:49:15 (lmgrd)   This log is intended for debug purposes only.

.....

9:49:15 (lmgrd)
9:49:15 (lmgrd) -----
9:49:15 (lmgrd)
9:49:15 (lmgrd)
9:49:15 (lmgrd) Server's System Date and Time: Mon Feb 28 2022 09:49:15 CET
9:49:15 (lmgrd) SLOG: Summary LOG statistics is enabled.
9:49:15 (lmgrd) FlexNet Licensing (v11.18.3.0 build 283040 x64_lsb) started on ↵
↵hostname (linux) (2/28/2022)
9:49:15 (lmgrd) Copyright (c) 1988-2021 Flexera. All Rights Reserved.
9:49:15 (lmgrd) World Wide Web:  http://www.flexerasoftware.com
9:49:15 (lmgrd) License file(s): /home/username/mosek/mosek-floating.lic
9:49:15 (lmgrd) lmgrd tcp-port 27007
9:49:15 (lmgrd) (@lmgrd-SLOG@) =====
9:49:15 (lmgrd) (@lmgrd-SLOG@) === LMGRD ===
9:49:15 (lmgrd) (@lmgrd-SLOG@) Start-Date: Mon Feb 28 2022 09:49:15 CET
9:49:15 (lmgrd) (@lmgrd-SLOG@) PID: 2147425
9:49:15 (lmgrd) (@lmgrd-SLOG@) LMGRD Version: v11.18.3.0 build 283040 x64_lsb ( build ↵
↵283040 (ipv6))
9:49:15 (lmgrd) (@lmgrd-SLOG@)
9:49:15 (lmgrd) (@lmgrd-SLOG@) === Network Info ===
9:49:15 (lmgrd) (@lmgrd-SLOG@) Listening port: 27007
9:49:15 (lmgrd) (@lmgrd-SLOG@)
9:49:15 (lmgrd) (@lmgrd-SLOG@) === Startup Info ===
9:49:15 (lmgrd) (@lmgrd-SLOG@) Server Configuration: Single Server
9:49:15 (lmgrd) (@lmgrd-SLOG@) Command-line options used at LS startup: -c /home/
↵username/mosek/mosek-floating.lic
9:49:15 (lmgrd) (@lmgrd-SLOG@) License file(s) used: /home/username/mosek/mosek-
↵floating.lic
9:49:15 (lmgrd) (@lmgrd-SLOG@) =====
9:49:15 (lmgrd) Starting vendor daemons ...
9:49:15 (lmgrd) Started MOSEKLM (internet tcp_port 36867 pid 2147427)
9:49:15 (MOSEKLM) FlexNet Licensing version v11.18.3.0 build 283040 x64_lsb
9:49:15 (MOSEKLM) SLOG: Summary LOG statistics is enabled.
9:49:15 (MOSEKLM) SLOG: FNPLS-INTERNAL-CKPT1
9:49:15 (MOSEKLM) SLOG: VM Status: 0
9:49:15 (MOSEKLM) SLOG: FNPLS-INTERNAL-CKPT5
9:49:15 (MOSEKLM) SLOG: TPM Status: 0
9:49:15 (MOSEKLM) SLOG: FNPLS-INTERNAL-CKPT6
9:49:15 (MOSEKLM) Server started on hostname for:          PTS
9:49:15 (MOSEKLM) PTON
9:49:15 (MOSEKLM) EXTERNAL FILTERS are OFF
9:49:15 (lmgrd) MOSEKLM using TCP-port 36867
9:49:15 (MOSEKLM) SLOG: Statistics Log Frequency is 240 minute(s).
9:49:15 (MOSEKLM) SLOG: TS update poll interval is 600 seconds.
9:49:15 (MOSEKLM) SLOG: Activation borrow reclaim percentage is 0.
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) =====
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) === Vendor Daemon ===
```

(continues on next page)

(continued from previous page)

```
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Vendor daemon: MOSEKLM
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Start-Date: Mon Feb 28 2022 09:49:15 CET
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) PID: 2147427
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) VD Version: v11.18.3.0 build 283040 x64_lsb (
↳build 283040 (ipv6))
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@)
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) === Startup/Restart Info ===
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Options file used: None
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Is vendor daemon a CVD: No
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Is FlexNet Licensing Service installed and
↳compatible: No
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) FlexNet Licensing Service Version: -NA-
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Is TS accessed: No
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) TS access time: -NA-
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Number of VD restarts since LS startup: 0
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@)
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) === Network Info ===
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Listening port: 36867
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Daemon select timeout (in seconds): 1
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@)
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) === Host Info ===
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Host used in license file: hostname
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) HostID node-locked in license file: b47acd81a33c
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) HostID of the License Server: "b47acd81a33c
↳b47acd81a33d"
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) Running on Hypervisor: Not determined - treat as
↳Physical
9:49:15 (MOSEKLM) (@MOSEKLM-SLOG@) =====
9:49:15 (MOSEKLM) TCP_NODELAY NOT enabled
9:49:15 (MOSEKLM) Listener Thread: running
9:49:15 (MOSEKLM) Starting diagnostics port listener thread (DPLT)
9:49:15 (MOSEKLM) Starting diagnostics output thread (DRQT)
```

In particular, the major information contained in the file is:

- the port `lmgrd` is running on (in this case 27007, the default port used by **MOSEK**),
- the port `MOSEKLM` is running on (in this case 36867, by default a value assigned dynamically at startup),
- the path to the license file that was used,
- the hostname and hostid of the machine,
- the licensed part IDs (in this case PTS and PTON)

If the log file indicates errors consult [Sec. 11.2](#).

7.2.3 Testing the Token Server

In order to verify that the token server is running, go to the `bin` folder of the **MOSEK** installation mentioned above and run:

```
./lmutil lmstat -c 27007@127.0.0.1 -a
```

(if necessary replace 27007 with your port number). The result for a successfully running token server should be similar to the one below:


```
lmutil - Copyright (c) 1989-2021 Flexera. All Rights Reserved.
Flexible License Manager status on Mon 2/28/2022 10:09

License server status: 27007@hostname
  License file(s) on hostname: /home/username/mosek/mosek-floating.lic:

  hostname: license server UP (MASTER) v11.18.3

Vendor daemon status (on hostname):

  MOSEKLM: UP v11.18.3
Feature usage info:

Users of PTS:  (Total of 5 licenses issued;  Total of 0 licenses in use)

Users of PTON: (Total of 5 licenses issued;  Total of 0 licenses in use)
```

7.2.4 Troubleshooting

If the server won't start, or the client can't check out the license first check [Sec. 11.2](#). It covers most typical issues. Always consult the log file `lmgrd.log` of the server.

If that doesn't help please contact **MOSEK** support at support@mosek.com. Please include the client error messages and the server log file `lmgrd.log`. See [Sec. 11.4](#).

If on Linux you get an error similar to:

```
./lmgrd: No such file or directory
```

then most likely the Linux Standard Base `lsb` package is not installed.

7.2.5 Starting lmgrd on boot

To start the license server automatically on boot add the above procedure to your system's startup scripts. Details vary depending on the operating system. Note that for security reasons `lmgrd` should not run as root, therefore we recommend a construction such as:

```
su USERNAME -c "umask 022; LMGRD -c PATH_TO_LICENSE_FILE -l PATH_TO_LOG_FILE"
```

where

- `USERNAME` is a normal, non-root, non-privileged user,
- `LMGRD` is the complete path to the `lmgrd` binary,
- `PATH_TO_LICENSE_FILE` is the complete path to the license file,
- `PATH_TO_LOG_FILE` is the complete path to the debug log file.

7.3 Changing default ports and firewall and antivirus issues

The token server consists of two daemons

- `lmgrd`: The token server daemon. By default it listens on port `27007`.
- `MOSEKLM`: A demon started by `lmgrd`. Opens its own port; its number can vary between runs unless explicitly specified (see below).

Both need an open port in the firewall if a **MOSEK** client should be able to check out a license license token. To specify explicitly which port number each daemon should use you must change the license file. The first two lines in a standard **MOSEK** floating license file look like

```
SERVER my_server 123456789ABC 27007
VENDOR MOSEKLM
```

To instruct `lmgrd` to use port `27008` and `MOSEKLM` to use port `3084` instead, change the first two lines of the license file to:

```
SERVER my_server 123456789ABC 27008
VENDOR MOSEKLM port=3084
```

Restart the token server and configure your firewall to allow access to the chosen port numbers, in this case `27008` and `3084`.

Finally, it is a good idea to check if the port is open by using the `telnet` command as follows

```
telnet my_server 27008
```

on the client computer(s). If you get an error message similar to

```
Connecting to my_server...Could not open connection to the host,
on port 27008: Connect failed
```

then the port is *not* open. See also the [License Administration Guide](#) for more information.

Note that antivirus software can have a similar effect and also block connections, even from the same machine. In this case instruct the software to allow connections through the two ports you specify explicitly in the license file as shown above.

7.4 Administration of the token server

We briefly describe how to stop, disable, restart the token server, update the license file and upgrade the server. A full administrator's manual with all options and details can be found in [License Administration Guide](#).

7.4.1 Using command line

If the `lmgrd` token server process was started using the command line (Linux, macOS or command-line on Windows) then all operations related to stopping/restarting/updating/upgrading can be carried out by stopping the `lmgrd` process and starting a new one.

To stop the token server gracefully, run

```
./lmutil lmdown -c PATH_TO_LICENSE_FILE
```

The program `lmutil` can be found in the `bin` folder inside the **MOSEK** installation, that is:

```
<MSKHOME>/mosek/10.0/tools/platform/<PLATFORM>/bin/
```

where `<PLATFORM>` is the platform you are working on.

Now you can start a new license server, if required, with the desired license file and configuration, from the same or another **MOSEK** installation, according to the instructions in [Sec. 7.2](#).

7.4.2 Using LMTOOLS

If the token server was started using `lmtools` then all operations related to stopping/restarting/updating/upgrading can be carried out through `lmtools`. Start `lmtools` from the `bin` folder of the **MOSEK** installation or from the **MOSEK** Start Menu.

Replacing the license file

- If you are changing to a new license file, go to the **Config services** TAB in **lmttools**, choose your license server from the list in *Service Name*, update the path in *Path to the license file* and click *Save Service*. This will automatically restart the license server using the new file.
- If you are overwriting the current file, then go to the **Start/Stop/Reread** TAB in **lmttools**, choose your license server from the list, and click *ReRead License File*.

Check if the restart succeeded by performing a status enquiry as in [Sec. 7.1.5](#) and if necessary use the troubleshooting instructions as for the installation process.

Stopping

To stop the server go to the **Start/Stop/Reread** TAB in **lmttools**, choose your license server from the list, and click *Stop Server*.

Starting

To start a previously stopped server go to the **Start/Stop/Reread** TAB in **lmttools**, choose your license server from the list, and click *Start Server*.

Removing the service

To remove the service from the system first stop the server if not already stopped (see above). Then go to the **Config services** TAB in **lmttools**, choose your license server from the list in *Service Name* and click *Remove Service*.

Upgrading the token server

To upgrade the token server to a new version first remove the existing service (see above) and then repeat the installation process [Sec. 7.1](#) using another version of **lmttools** (for instance from another installation of **MOSEK**).

Chapter 8

Client setup

This section describes setting up client machines.

8.1 With a local license file

Default setup

The preferred option is to place the license file `mosek.lic` in the directory `mosek` in the user's home directory. That means

```
$HOME/mosek/mosek.lic  
%USERPROFILE%\mosek\mosek.lic
```

on UNIX systems and Windows, respectively. If no other configuration options are set (see below) this is the default location where **MOSEK** looks for a license. This works for all types of licenses. If the license file contains a floating license, the client will use the information in that file to find and contact a token server.

Environment variable

Alternatively, the path to the license file may be set by the environment variable `MOSEKLM_LICENSE_FILE`, for example:

```
MOSEKLM_LICENSE_FILE=/home/user/licenses/mosek.lic  
MOSEKLM_LICENSE_FILE=c:\users\mylogin\licenses\mosek.lic
```

Command line options

From the **MOSEK** command line the path to the license file can be set with the option `-l`.

From an API

In the Optimizer API and Fusion API the path to the license file can be set with the method `putlicensepath` of the environment or Model, respectively, before first optimization.

8.2 Without a local license file

Another method to check out a license from a floating license token server is to set the environment variable `MOSEKLM_LICENSE_FILE` in one of the following formats

```
MOSEKLM_LICENSE_FILE=@hostname  
MOSEKLM_LICENSE_FILE=port@hostname
```

where `hostname` is the name of the token server machine and `port` is the port on which `MOSEKLM` is listening. Then the client **MOSEK** application will contact the server directly and the possible overhead for opening and reading the license file is eliminated. Observe a potential firewall may block access to the token server.

The same format can be used in conjunction with the command line `-l` option and in the `putlicensepath` method as mentioned in the previous section.

8.3 Testing and debugging

In either case the client configuration can be tested by running the program `msktestlic`.

If any errors related to licensing appear, then go through the error messages and look for a line containing the path to the license file, for example:

```
License path: /home/someplace/mosek/mosek.lic
```

or:

```
License cannot be located. The default search path is ':/home/someplace/mosek/mosek.  
↪lic:'
```

This is the *actual* location where this instance of **MOSEK** was trying to locate the license. If it does not correspond to your expectations then go through the setup again or check that this location contains the correct file.

Chapter 9

Additional features

Here we describe some frequently requested additional features available for **floating licenses**.

9.1 Options file

FLEXlm allows some customizations by the user through an options file. An options file (say, `mosek.opt`) can be attached to the license by editing the `VENDOR` line in the license file as follows:

```
VENDOR MOSEKLM OPTIONS=mosek.opt
```

The options file is a text file with one option per line. The full list of options can be found in the section *Managing the options file* of the [FLEXnet license administration guide](#). Some useful ones are:

- **MAX**: restricts the number of tokens one user/group can check out. For example, to restrict user `bill` to at most 2 tokens of each type `PTS` and `PTON` use the option file with content:

```
MAX 2 PTS USER bill
MAX 2 PTON USER bill
```

Replacing `bill` with `ALL_USERS` will introduce this restriction for all users. Apart from `USER`, such restrictions can also be applied to `HOST` (hostname), `INTERNET` (IP address) and other classifiers.

- **RESERVE**: reserves a number of tokens for one user/group. For example, to reserve one token of each type `PTS` and `PTON` for user `bill` use the option file with content:

```
RESERVE 1 PTS USER bill
RESERVE 1 PTON USER bill
```

Apart from `USER`, such restrictions can also be applied to `HOST` (hostname), `INTERNET` (IP address) and other classifiers.

Not all options provided by FLEXlm are enabled in **MOSEK**. If the options file was processed successfully, relevant messages will appear in the license server's log file during startup, for example:

```
15:08:30 (MOSEKLM) Using options file: "mosek.opt"
...
...
15:08:30 (MOSEKLM) MAX 2 USER bill for PTS
15:08:30 (MOSEKLM) MAX 2 USER bill for PTON
```

9.2 Borrowing a license

The `lmborrow` functionality allows to check out a license from the token server and keep it checked out (borrowed) on another machine for a specified period of time, without a connection to the token server. This way **MOSEK** can be used on a computer disconnected from the network, for example a laptop used for field service work. When the borrowing period ends, the license automatically expires on the borrowing machine and returns to the token pool of the token server.

Configuring a borrowed license involves a few steps. We assume that a **MOSEK** floating license is configured and working in advance.

Initialize borrowing

Initially the borrowing client must be connected to the server and must be able to check out a license in a normal way.

- From the bin of **MOSEK** on the client machine run

```
lmutil lmborrow MOSEKLM 20-may-2023 15:00
```

The date (required) and time (optional) specify the **end** of the borrowing period.

- Run some code which requires **MOSEK** on the borrowing client in order to check the out the licenses for the first time. Make sure to run code which checks out all the parts (PTS and if needed PTON) you intend to borrow. You can verify that the required licenses were borrowed by running

```
lmutil lmborrow -status
```

It should produce output similar to

```
lmutil - Copyright (c) 1989-2021 Flexera. All Rights Reserved.
Vendor      Feature                               Version    Expiration
-----
MOSEKLM     PTS                               10.00      20-may-23 15:00
MOSEKLM     PTON                              10.00      20-may-23 15:00
```

Using the borrowed license

You can now disconnect the client and use the borrowed license in offline mode until its expiry time. When the borrowing period expires the token server will reclaim the borrowed license automatically.

Early check in

You can return the borrowed license to the token server before its designated expiry time. To do this, connect the borrowing client back to the network so that it can communicate with the token server and, depending on which features you borrowed, run one or both of the commands:

```
lmutil lmborrow -return PTS
lmutil lmborrow -return PTON
```

Chapter 10

License in a Cloud Computing Environment

The token server may be deployed in a cloud environment. The main challenge in deploying a token server in the cloud is to guarantee that the `hostid` (in this case the MAC address) stays unchanged when the instance running the token server is stopped.

In the following section we discuss one possible deployment strategy on Amazon EC2.

10.1 Example: Token server in Amazon EC2

The license will be bound to a MAC address. In the most basic Amazon EC2 instance setup the MAC address may change when the instance is stopped and later started again. Below we describe how to work around this.

In Amazon EC2 a MAC address is a persistent resource associated with an Elastic Network Interface (ENI). To keep the MAC address constant we advise creating an ENI that can then be associated with the Amazon EC2 instance acting as a token server. The ENI can later be moved to another instance within the same subnet if the token server needs to be moved to another instance.

Creating a token server Amazon EC2 instance

1. Create an ENI in the subnet into which you wish to launch the token server. Please consult the Amazon EC2 documentation for how to create an ENI.
2. Create a new instance in the same subnet as the ENI. When configuring the network interface select the newly created ENI as a network interface.
3. Launch the instance.
4. (*optional*) If the machine needs a public IP address then create an Elastic IP (EIP) and associate it with the instance after launch. It is not possible to have an automatically assigned public IP addresses when using an ENI in Amazon EC2.
5. Install **MOSEK** on the instance.
6. Retrieve the MAC address associated with the ENI, e.g by inspecting the ENI in the AWS Management Console or by logging into the instance and following the instructions in [Sec. 6](#).
7. Contact support@mosek.com with the relevant MAC address to obtain a valid license file.
8. Make sure the security group associated with the instance running the token server allows for incoming traffic to the token server. Allow for inbound TCP traffic on the ports your token server is listening as shown in [Sec. 7.3](#)
9. Install the token server as described in [Sec. 7](#).

Chapter 11

Licensing FAQ

11.1 General questions

11.1.1 How do I know what version/expiry/hostid/port I have in my license file?

The license file is a plain text file. Open it in any text editor and follow [Sec. 5.2](#).

11.1.2 Can I edit the license file?

You can change the hostname, port number(s), add an `OPTIONS` section and edit comments. Any other edits (in particular changing the hostid) will invalidate the license file.

11.1.3 Can I reserve some of my floating tokens to a specific IP/username/group?

To reserve a certain number license features for a particular user or IP address, you must create an additional options file and use the `RESERVE` option. For details see [Sec. 9.1](#). For example, to reserve one PTS token for user `username` make an options file `res.opt` with content

```
RESERVE 1 PTS USER username
```

and add the following to your license file:

```
VENDOR MOSEKLM OPTIONS=res.opt
```

11.1.4 Can I check out licenses from more than one location?

You can concatenate multiple license search paths (local or remote) with the operating system's path separator. **MOSEK** will then sequentially try to check out a license from all these locations until one is available. For example

<code>MOSEKLM_LICENSE_FILE=/home/user/licenses/mosek.lic:27007@licensehost</code>	(Linux/ → OSX)
<code>MOSEKLM_LICENSE_FILE=c:\users\mylogin\licenses\mosek.lic;27007@licensehost</code>	(Windows)

11.1.5 Can I use more than one hostid?

Yes, for a server license. In exceptional situations we can generate a server license file with a list of MAC addresses as `hostid`. At least one of them has to be active at any time for **MOSEK** to work. For a floating license we can use only a single `hostid`.

11.1.6 My machine has many network interfaces and I get a long list of `hostids`. Which one should I use?

Ideally choose the MAC address of some permanent physical device (network card) that is expected to be permanently associated with the machine. Avoid MAC addresses of temporary/virtual devices/interfaces which can easily disappear, rendering the license file invalid.

11.1.7 Is it possible to use **MOSEK** with a floating license on a machine detached from the LAN(WAN)?

Yes. It is possible to use the `lmborrow` functionality. See the [Sec. 9.2](#).

11.1.8 Is it possible to host a token server in the cloud?

Yes, but you will have to create a (small) instance with a permanent MAC address to be used as `hostid`. An example is described in [Sec. 10](#).

11.1.9 Do you have a complete licensing guide that covers all of **FLEXlm** options?

The ultimate guide is at [FLEXnet license administration guide](#).

11.1.10 I know the `hostid` of the license server, but what about the clients?

The clients, that is machines on which **MOSEK** optimizations will actually be performed, can be arbitrary, as long as they can connect to the license server to check out a token. The floating license restricts only the physical machine for the token server.

11.1.11 What does **MOSEK** use to verify the license?

Only the license file. There is no external communication whatsoever.

11.1.12 When does **MOSEK** return the floating license token?

By default at the end of the session/process or when the **MOSEK** environment is deleted. This reduces the checkout overhead if one process runs multiple optimizations. However, this is completely customizable. For example, you can return the token after each optimization or upon calling a dedicated method. Your API's manual has instructions on how to do it in the section *Technical guidelines / The license system*.

11.2 Floating license issues

11.2.1 The `lmutil` and `lmgrd` will not start on Linux: No such file or directory.

If you run a command from the licensing system and get this error:

```
user@hostname:~/path_to_mosek$ ./lmutil
./lmutil: No such file or directory
```

then most likely you are missing the Linux Standard Base (LSB) package. LSB version at least 3 is required. On Ubuntu the latest version of LSB can be installed with the command `apt-get install lsb`.

11.2.2 I get the error `Vendor daemon is too old`.

You are running an older license server (for example, from version 7) and tried to connect to it with a much more recent client (for example, from version 9). The `lmgrd` license server must be at least as new as all the clients who attempt to connect to it. Use the license server distributed with the most recent version of the clients in use. We occasionally update the FLEXlm binaries inside **MOSEK** and then an incompatibility with older license servers may occur.

11.2.3 The license server will not start.

Always check the server's log file `lmgrd.log`. It will explain the source of the problem. Typical issues are:

- **Wrong hostname:**

```
11:55:03 (lmgrd) "HOSTNAME_REAL": Not a valid server hostname, exiting.
11:55:03 (lmgrd) Valid license server system hosts are: "HOSTNAME_LICFILE"
```

The hostname in the license file is not the actual hostname of the computer. Either you are running the license server on the wrong machine, or using a wrong file or something changed on the machine and you should contact us for a new license file.

- **Wrong hostid:**

```
8:46:36 (MOSEKLM) (@MOSEKLM-SLOG@) HostID node-locked in license file:
→a6321bc7ff3f
8:46:36 (MOSEKLM) (@MOSEKLM-SLOG@) HostID of the License Server: "af638abf82aa
→127c4da8f212"
8:46:36 (MOSEKLM) No valid hostids, exiting
8:46:36 (MOSEKLM) EXITING DUE TO SIGNAL 25 Exit reason 2
8:46:41 (lmgrd) MOSEKLM exited with status 25 (Invalid host)
```

The hostid in the license file does not correspond to actual hardware. Either you are running the license server on the wrong machine, or using a wrong file or something changed on the machine (perhaps a network interface was removed) and you should contact us for a new license file.

- **Incorrect or unreadable license file, for example:**

```
18:54:46 (MOSEKLM) Invalid license key (inconsistent authentication code)
```

It is possible that you invalidated the file by editing hostname or hostid or that you downloaded/saved it incorrectly. See [Sec. 5.2](#) for how a floating license file should look. You should revert to the original file (if applicable) or contact us for a new license file, attaching the current one.

- **Could not open port:**

```
18:52:12 (lmgrd) Failed to open the TCP port number in the license.
```

The token server or vendor daemon could not open a port. Possibly it is already open by another process, or there are firewall issues. In either case consult [Sec. 7.3](#) for a solution. Terminate any `lmgrd` or `MOSEKLM` processes running.

- **Invalid License File**, on Unix:

```
18:54:46 (lmgrd) Server's System Date and Time: Mon Jun 21 2021 18:45:01 CET
18:54:46 (lmgrd) Invalid License File
```

Can appear if the license file has permissions other than 644. Change the license file's permissions to 644.

- **Hostname not in network database:**

```
11:55:03 (lmgrd) Unknown Hostname: HOSTNAME_LICFILE
specified in the license file is not available in the local network database
11:55:03 (lmgrd) EXITING DUE TO SIGNAL 33 Exit reason 1
```

A rare error appearing on MAC where `lmgrd` exits almost immediately. The solution is to edit the file `/etc/hosts` in your system and add a line with the IP address and hostname of your machine.

11.2.4 The client cannot connect to the license server or license checkout times out.

The following symptoms can indicate firewall issues:

- The client cannot connect to the license server. The error is `MOSEK cannot connect to the license server` or `Operation now in progress`.
- The client cannot connect to the license server from a remote machine, but everything works when the client runs on the same machine as the server.
- License checkout times out after about 60 seconds with message `Timeout: operation now in progress`.

First make sure that the license server is up and running and that the `lmgrd.log` file did not indicate any errors (see previous question). Assuming that, consult [Sec. 7.3](#), fix *both ports* described in that section (the token server port and the vendor port) to your preferred values, make sure they are open in the firewall and/or antivirus software, restart the license server and try again.

This happens typically in networks with tight security settings. Occasionally antivirus software can trigger this problem.

11.2.5 I cannot check out a token even though there should be more available.

Most likely there are processes/users who checked out a license but never returned it. The `lmgrd.log` log file on the license server will contain the history of license ckeckouts and checkins which may be used to identify the culprit.

11.2.6 The floating license worked previously and now suddenly stopped.

The most typical reasons for this are:

- The token server was terminated.
- The `hostid` of the license server changed.
- Something changed in the network configuration, for example there are new firewall or antivirus settings preventing the connection.
- The server was restarted and now uses a different port or different vendor port.
- You are trying to check out too many tokens.

Consult the error message reported by the **MOSEK** client for more information. Make sure that the license server is up and running and that the `lmgrd.log` file did not indicate any errors (see previous questions).

If the symptoms resemble [Sec. 11.2.4](#), which indicates firewall issues, then consult [Sec. 7.3](#), fix *both ports* described in that section (the token server port and the vendor port) to your preferred values, make sure they are open in the firewall and/or antivirus software, restart the license server and try again.

If the reason is a change in `hostid` contact us for a new license file.

11.3 Local file issues

11.3.1 The Optimization Toolbox for Matlab or Rmosek says license has expired although I downloaded a new one.

After you put the license file in the right place restart Matlab or R. It caches the license. If that doesn't help **MOSEK** probably uses another license path than you first thought. See next question.

11.3.2 MOSEK says there is no license file or that it expired.

If you are sure that your license file is valid and in the right place, then it is possible that your system is configured to expect the license file elsewhere than you thought. Search for the `mosek.lic` file on the disk and see if there is one in another location. Remove any old expired licenses found.

Go through the error or diagnostic message you received from **MOSEK** and look for lines such as:

```
License path: /home/someplace/mosek/mosek.lic
```

or:

```
License cannot be located. The default search path is ':/home/someplace/mosek/mosek.  
↪lic:'
```

This is the *actual* location where this instance of **MOSEK** was trying to locate the license. If it does not correspond to your expectations then go through the setup again or check that this location contains the correct file.

11.4 Need more help

11.4.1 I tried all of the above but it doesn't help.

Send us an email and attach as much as possible of the following data:

- description of the problem,
- the license file,
- the token server log file `lmgrd.log` or its part relevant to the issue (for floating licenses),
- the exact error message reported by the **MOSEK** client, or the output from `mosekdiag`, `mosektestlic`, `mosek -f` or other relevant diagnostic tool.
- **MOSEK** version, operating system, which interface you are using.

Index

F

floating license, [11](#)

I

installation |linux|
 token server, [27](#)

installation |macos|
 token server, [27](#)

installation Windows
 token server, [20](#)

N

node-locked license, [11](#)

S

server license, [11](#)

T

token server, [19](#)
 installation |linux|, [27](#)
 installation |macos|, [27](#)
 installation Windows, [20](#)